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United States Patent	12403974
Kind Code	B2
Date of Patent	September 02, 2025
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Electric scooter

Abstract

Various personal mobility vehicles, such as scooters, are disclosed. The scooter can include at least one battery and motor for powering at least one driven wheel. The vehicle can include a gear assembly to convert a torque produced by the motor to a different torque to a driven shaft to power the at least one driven wheel. The battery can be removable.

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Appl. No.: 18/656005

Filed: May 06, 2024

Prior Publication Data

Document Identifier	Publication Date
US 20240359763 A1	Oct. 31, 2024

Related U.S. Application Data

continuation parent-doc US 16263801 20190131 US 11975794 child-doc US 18656005
continuation parent-doc US 15409488 20170118 ABANDONED child-doc US 16263801
continuation parent-doc US 29577038 20160908 US D818541 20180522 child-doc US 16263801
us-provisional-application US 62286124 20160122
us-provisional-application US 62340978 20160524

Publication Classification

Int. Cl.: **B62J43/13** (20200101); **B60K1/04** (20190101); **B60K7/00** (20060101); **B60K17/04** (20060101); **B60K17/26** (20060101); **B62J43/28** (20200101); **B62K3/00** (20060101); **B62K15/00** (20060101); **B62K19/40** (20060101); **B62M11/02** (20060101); B62J50/22 (20200101)

U.S. Cl.:

CPC **B62J43/13** (20200201); **B60K1/04** (20130101); **B60K7/0007** (20130101); **B60K17/043** (20130101); **B60K17/26** (20130101); **B62J43/28** (20200201); **B62K3/002** (20130101); **B62K15/006** (20130101); **B62K19/40** (20130101); **B62M11/02** (20130101); B60K2001/0438 (20130101); B60K2001/0455 (20130101); B60K2001/0483 (20130101); B60K2001/0494 (20130101); B60K2007/0046 (20130101); B60K2007/0061 (20130101); B60Y2200/126 (20130101); B62J50/22 (20200201); B62K2204/00 (20130101)

Field of Classification Search

CPC: B62J (43/13); B62J (43/28); B60K (1/04); B60K (7/0007); B60K (2001/0438); B60K (2001/0455); B60Y (2200/126)

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Background/Summary

CROSS REFERENCE (1) This application is a continuation of U.S. patent application Ser. No. 16/263,801, filed Jan. 31, 2019, now U.S. Pat. No. 11,975,794, which is a continuation of U.S. patent application Ser. No. 15/409,488, filed Jan. 18, 2017, which claims the priority benefit under at least 35 U.S.C. § 119 of U.S. Patent Application No. 62/286,124, filed Jan. 22, 2016, and U.S. Patent Application No. 62/340,978, filed May 24, 2016, and which is a continuation in part of U.S. patent application Ser. No. 29/577,038, filed Sep. 8, 2016, now Pat. No. D818,541. The entirety of each of the aforementioned applications is incorporated by reference herein.

BACKGROUND

Field

(1) Embodiments of the disclosure relate generally to personal mobility vehicles, such as electric scooters that can freewheel in at least one direction.

Description of Certain Related Art

(2) The use of personal mobility vehicles, such as scooters, has become a popular recreational activity as well as useful means of personal transportation. Scooters have become popular among many age groups and there are many different scooter variations and scooter designs. Conventional scooters have at least two wheels and a steering mechanism. Scooter designs have become more compact and convenient, and some scooters have folding mechanisms allowing the user to easily store or carry the scooter. In order to allow a user to slow or stop a scooter, some scooters include a braking mechanism.

SUMMARY OF CERTAIN FEATURES

(3) With the increased popularity of scooters, variations of the conventional scooters have gained popularity as well, including motorized scooters. With the use of electric or motorized scooters, the

weight of an electric motor may limit the portability of the scooter, particularly for commuters. Additionally, use of large electric motors can require large batteries, further increasing the weight of the scooter. Thus there is a need to provide an electric scooter having a small motor and battery to allow for a lightweight and portable electric scooter at a low cost.

(4) Some electric powered scooter embodiments include an electric motor mounted to the front or rear of the scooter frame near at least one of the front or rear wheels, with a drive assembly coupled to the motor and configured to transfer drive power from the motor to the driven front or rear wheel. The drive assembly can include powered rollers using frictional directional force upon the circumference of the driven wheel or a drive chain linking the drive means to the hub of the driven wheel. The driven wheel can be electrically driven when a throttle assembly controlling the motor, commonly located on or about the handlebar, is activated by the user. Some electric powered scooter embodiments include a battery compartment accessible when the scooter is a folded configuration.

(5) Scooters having a driven wheel that is not able to freewheel in one direction can have several disadvantages, such as unwanted noise, undesirable friction, and/or difficulty moving or transporting the scooter due to the wheel being unable to move unless powered. Certain of the electric powered scooter embodiments disclosed herein address these issues, or other issues, by allowing the driven wheel to freewheel in at least one direction so that the scooter may be moved when unpowered.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) These and other features, aspects, and advantages are described below with reference to drawings of an example embodiment, which are intended to illustrate, but not to limit, the present disclosure.
- (2) FIG. 1 is a perspective view of an embodiment of a scooter.
- (3) FIG. 2 is a close up view of a wireless remote throttle that may be used with the scooter of FIG. 1.
- (4) FIG. 3 is a rear perspective view of the scooter of FIG. 1.
- (5) FIG. 4 is a close up view of a display and throttle arrangement of the scooter of FIG. 1.
- (6) FIG. 5 is a close up rear perspective view of a first side of the rear wheel assembly of the scooter of FIG. 1.
- (7) FIG. 6 is a partial exploded view of the rear wheel assembly of the scooter of FIG. 1.
- (8) FIG. 7 is a close up rear perspective view of a second side of the rear wheel assembly of the scooter of FIG. 1.
- (9) FIG. 8 is a close up exploded view of the rear wheel assembly of the scooter of FIG. 1.
- (10) FIG. 9 is another exploded view of the rear wheel assembly of the scooter of FIG. 1.
- (11) FIG. 10 is another exploded view of the rear wheel assembly of the scooter of FIG. 1.
- (12) FIG. 11 is a perspective view of the scooter of FIG. 1 in a folded configuration.
- (13) FIG. 12 is a perspective view of the scooter of FIG. 11 with a battery pack removed.
- (14) FIG. 13 is a partial cross-sectional view of the scooter of FIG. 11.
- (15) FIG. 14 is a close up view of the cross-sectional view of FIG. 13.
- (16) FIG. 15 is a perspective view of another embodiment of a scooter.
- (17) FIG. 16 is a front view of the scooter of FIG. 15.
- (18) FIG. 17 is a rear view of the scooter of FIG. 15.
- (19) FIG. 18 is a left-side view of the scooter of FIG. 15.
- (20) FIG. 19 is a right-side view of the scooter of FIG. 15.
- (21) FIG. 20 is a top view of the scooter of FIG. 15.

(22) FIG. 21 is a bottom view of the scooter of FIG. 15.

DETAILED DESCRIPTION OF CERTAIN EMBODIMENTS

(23) Reference will now be made in detail to various embodiments of the present technology, which relates to various personal mobility devices, such as scooters. Although certain specific embodiments of the present technology are described, the present technology is not limited to these embodiments. On the contrary, these described embodiments are merely illustrative of the present technology, and the present technology is intended to also cover alternatives, modifications, and equivalents. Furthermore, in the following detailed description, numerous specific details are set forth in order to provide a thorough understanding of the present technology. However, it will be recognized by one of ordinary skill in the art that embodiments may be practiced without these specific details. In some instances, well known methods, procedures, compounds, compositions and mechanisms have not been described in detail as not to unnecessarily obscure aspects of embodiments of the present technology.

(24) FIGS. 1-14

(25) FIGS. 1-14 illustrate an embodiment of an electric scooter 10. As shown in FIG. 1, the scooter 10 includes a scooter body 20, a support surface 34, a rotatable shaft 32, and a handlebar assembly 24. The support surface 34 can be configured to support at least one foot of the user. The illustrated support surface 34 is configured to accommodate both feet of a user, such as in one or both of a fore-and-aft or side-by-side arrangement. The scooter 10 includes at least one front wheel 28 and at least one rear wheel 30 connected with the scooter body 20. The front wheel 28 and the rear wheel 30 can be spaced apart from one another with the support surface 34 extending therebetween. The wheels 28, 30 can be aligned in the same plane, as depicted, and located at opposite ends of the support surface 34. As depicted, some embodiments of the scooter 10 include a rotatable shaft 32 which rotates with the handlebar assembly 24 so that the handlebar assembly 24 can rotate or swivel within a head tube of the scooter body 20. The shaft 32 can be connected with a fork 31 that connects with the front wheel 28, such that the handlebar assembly 24 can be rotated to swivel the front wheel 28 to steer the scooter 10. The fork 31 can be directly connected to sides of the front wheel 28 and/or an axle that extends through a rotational axis of the front wheel 28.

(26) In some embodiments, the scooter 10 includes a folding assembly 70 and/or a motor assembly, such as a rear drive assembly 80. The folding assembly 70 allows the handlebar assembly 24 and the rotatable shaft 32 to pivot relative to the body 20 such that the handlebar assembly 24 folds against the body 20 for compact transport and storage of the scooter 10. Accordingly, in some embodiments, the rotatable shaft 32 can pivot about the folding assembly 70.

(27) The rear drive assembly 80 can include a motor 93 (see FIG. 6) to drive the rear wheel 30. In certain embodiments, torque from the motor 93 is transmitted to the wheel 30 via a transmission, such as a gear set or gear assembly. The folding assembly 70 and the rear drive assembly 80 are discussed in greater detail below.

(28) As shown in FIG. 2, in some embodiments, the handlebar assembly 24 includes a control mechanism, such as a wireless throttle remote assembly 26. The wireless throttle remote assembly 26 includes a throttle 60 that can be toggled by the user to increase or decrease the speed of a motor 93 to increase or decrease the speed of the scooter. In some embodiments, the throttle assembly 26 is wireless; however, in other embodiments, the throttle assembly 26 is wired to a motor 93 and/or a battery. In some embodiments, the throttle assembly 26 may include a brake lever (not shown) to activate a braking assembly to slow or stop the scooter 10. The throttle assembly 26 can be connected to the braking assembly through a wireless and/or mechanical connection. In some embodiments, the throttle 60 comprises a button, switch, lever, or other actuator available to the hand of the user while the user operates the scooter 10. Depressing the button throttle 60 initiates a signal (e.g., an electric or RF signal) to a controller, which signals for power to be transferred from the battery to the electric motor 93, resulting in rotational power being transferred from the electric motor 93 to the at least one driven wheel. In certain embodiments, the user is able to engage the

throttle by a hand crank throttle mechanism mounted on the handlebar assembly. In some embodiments, the user is able to engage the throttle by a hand lever mechanism, or other similar hand actuated mechanisms.

(29) With reference to FIGS. 3 and 4, in some embodiments, the throttle assembly **26** includes an on/off switch or button **62** and/or a display **64**. The on/off switch **62** may be used to send a “power on” or “power off” signal to the battery and/or the motor **93**. The display **64** provides a visual indication of a characteristic of the scooter, such as battery level, speed, distance, direction, or other information. In some embodiments, the display **64** indicates the amount of battery power remaining. The display **64** may indicate the battery power as a digital percentage display, or as a color gradient (such as green, yellow, red), or with any other visual indicator. The display **64** may be configured to indicate when the battery should be replaced and/or recharged, or the amount of time remaining until the battery is fully discharged.

(30) FIG. 5 illustrates a close-up view of the rear wheel **30** and rear drive assembly **80** of the scooter **10**. A side piece **90** is removable to provide access to the rear wheel **30** in case the rear wheel should need repair or replacement, such as due to wear, etc. A side piece **90** can be connected to the rear drive assembly **80** to surround at least a portion of the rear drive assembly **80**. For example, as shown, the side piece **90** can include one or more contour features, such as ribs or fins. The side piece **90** can be connected to the rear drive assembly by a mechanical fastener and/or a snap-fit configuration. As also shown, the motor **93** can be contained in a protective shell, such as a generally cylindrical casing **97**. In some embodiments, the side piece **90** and the casing **97** form a unitary and/or integral unit. In other embodiments, the cylindrical casing **97** is separate from the side piece **90**.

(31) A close-up partial exploded view of the rear drive assembly **80** is illustrated in FIG. 6. The side piece **90** is removed to illustrate how various components connect together to provide power to rear wheel **30**. In some embodiments, the rear drive assembly **80** includes a gear **84** that connects to (e.g., is received in) a rotation connector **82**. The gear **84** and connector **82** are received (e.g., wedged) within the rear wheel **30**. In some configurations, this can provide a friction fit to secure the gear **84** and connector **82** within the rear wheel **30**.

(32) In some embodiments, the rotation connector **82** comprises a one-way bearing connector. The one-way bearing connector **82** can be configured to transmit torque in one direction and allow free motion in the opposite direction. In certain implementations, the one-way bearing connector **82** allows the wheel **30** to spin freely (also called “freewheeling”) in a one rotational direction and can be driven in an opposite rotational direction. For example, relative to the orientation of the scooter **10**, the one-way bearing connector **82** can enable the wheel **30** to freewheel when the scooter **10** is moved backward and to be driven to move the scooter **10** forward. In some embodiments, the wheel **30** is configured to rotate about a wheel axis **88** and/or to be driven by the motor **93**, such as through gearing. In some implementations, the one-way bearing **84** reduces or eliminates undesirable noises, such as clicking noises commonly heard with bearings, by allowing the rear wheel **30** to freewheel. The one-way bearing connector **82** fits within a connector receiving area **86** of a gear box housing **87**. The gear box housing **87** can house and/or surround at least one or more components of the rear drive assembly **80**. In some embodiments, the rear box housing **87** can connect directly and/or indirectly to the side piece **90** to surround at least a portion of the rear drive assembly **80**. The rear drive assembly **80** can include a plurality of bearings **81**, **83**, and **85**.

(33) In some embodiments, the rear drive assembly **80** includes a brake mechanism **96**. The brake mechanism **96** may be in the form of a lever, as shown in FIG. 6, which can be activated by a user's foot to apply pressure to the rear wheel **30** and result in a braking force due to friction between the rear wheel **30** and the brake mechanism **96**. The brake mechanism **96** may be formed integrally with the main structure of the rear deck or body of the scooter **10** or may be a separate component coupled with the main structure of the rear deck or body of the scooter **10**. In some embodiments, the brake mechanism **96** is similar or identical to the brake assembly disclosed in U.S. Pat. No.

8,813,892, which is hereby incorporated by reference in its entirety. In some embodiments, the brake mechanism **96** comprises a rear end that is configured to engage with the rear wheel **30** and/or a front end that is configured to engage with the casing **97**. The brake mechanism **96** can be biased by a biasing member, such as a spring. For example, the rear end of the brake mechanism **96** can be biased away from the rear wheel **30** and/or the front end of the brake mechanism **96** can be biased into engagement with a stop structure, such as the casing **97**. As shown in FIG. **1**, for example, the brake mechanism **96** can extend rearwardly from the main structure of the rear deck or body of the scooter **10** and can extend over at least a portion of the rear wheel **30**.

(34) FIG. **7** illustrates a view of the opposite or gear box side of the rear drive assembly **80** and FIG. **8** illustrates an exploded view of the side of the rear drive assembly **80** shown in FIG. **7**. A gear set **89** is housed within the gear box housing **87**. The gear box housing **87** may be part of the main structure of the rear deck or body of the scooter **10**. A portion of the gear set **89** can be enclosed on the outside with a gear box side piece **92** and a cover piece **91**. The side piece **92** can include one or more apertures, such as to allow airflow from ambient to the motor **93** for cooling. As shown, in some embodiments, the side piece comprises a stepped configuration, such as a first step **93a**, a second step **93b**, a third step **93c**, and a fourth step **93d**. Each of the steps can have a circular peripheral shape. In some embodiments, as illustrated, the first, second, and third steps **93a-93c** can have respective outer surfaces that are laterally offset (e.g., non-coplanar) from each other. In some implementations, the outer surfaces of the first and fourth steps **93a**, **93d** are substantially co-planar. In some embodiments, the second step **93b** includes at least one fin that extends forwardly from the second step **93b** to the first step **93a**. The gear set **89** can include a plurality of gears (e.g., a worm gear, spur gear, etc.) that can be used to convert the torque produced by the motor **93** into a torque that is used to power the rear wheel **30**. The motor **93** may be housed within the gear box **87** at or near the end of the deck or body of the scooter **10**. As illustrated, the gear box **87**, in some configurations, forms a part of the structure of the rear deck or tail of the scooter **10**. In some embodiments, the motor **93** may be a hub motor similar to those described in U.S. Patent Publication No. 2015/0133253, which is incorporated by reference herein in its entirety. In certain embodiments, the motor **93** may be a small toy motor, such as those used to drive electric motor-driven toys.

(35) As shown in FIGS. **8-10**, the motor **93** drives a motor shaft **94**, such as a driving shaft. In the embodiment illustrated, a gear **95a** of the gear set **89** is mounted on the shaft **94** and a gear **95e** of the gear set **89** is mounted on the rear wheel axis **88**. The gears **95a**, **95e** can be operably connected through intermediate gears, such as gears **95b**, **95c**, **95d**. Through the sequence of gears **95a-95e**, the torque of the motor **93** is reduced to drive the rear wheel via the rear wheel axis **88**. As discussed above, the gear **84** and bearing connector **82** allow the rear wheel **30** to be powered in a first, or forward direction, and allow the rear wheel **30** to freewheel in a second, or rearward direction. An exploded view of the rear drive assembly **80** is illustrated in FIGS. **9** and **10**. In various embodiments, the rear wheel **30** can freewheel with no or substantially no sound (e.g., clicking) that is detectable by a user in normal operation of the scooter. For example, in some embodiments, during freewheeling, the wheel **30** and/or the connector **82** emits a sound that is less than or equal to about 20 decibels.

(36) In some embodiments, the gear **84** has a diameter that is wide enough to be received within a receiving area of the bearing connector **82** and within the receiving area **86** of the gear box housing **87**, as can be seen in at least FIG. **6**. The gear **84** has a plurality of flanges **84a** extending radially outward from the outer surface of the gear **84**. The flanges **84a** are configured to fit within corresponding recesses **84b** located on the inside surface of the bearing connector **82**. Similarly, the inside surface of the receiving area **86** of the gear box housing has recesses **84c** that are also configured to receive the flanges **84a**. The length **L** of the gear **84** is larger than the length **L'** of the bearing connector **82** such that at least a portion of the gear **84** is received within the bearing connector **82** and at least a portion of the gear **84** is received within the receiving area **86** of the

gear box housing **87**.

(37) Rotation of the rear wheel axis **88** acts on the bearings **83**, **85**, which may be press fit or otherwise coupled to the axis **88**. The bearings **83**, **85** can be coupled with the gear **84** through a friction fit or other coupling mechanism such that rotation of the axis **88** also rotates the gear **84**. Rotation of the gear **84** in a first rotational direction induces rotation of the bearing connector **82** through the interaction of the flanges **84a** on the recesses **84b** and rotates the rear wheel **30** in the first rotational or driven direction. As discussed above, the bearing connector **82** also permits freewheel rotation of the rear wheel **30** in a second rotational direction opposite the first rotational direction when the motor **93** is not powering the rear wheel **30** through the rear drive assembly **80**, as discussed above.

(38) FIGS. **11-13** illustrate the scooter **10** in a folded configuration. As discussed above, the folding assembly **70** allows the handlebar assembly **24** to pivot relative to the body **20** such that the rotatable shaft **32** and the handlebar assembly **24** fold against to the body **20** for compact transport and storage of the scooter **10**. As shown in FIGS. **11-13**, when the scooter **10** is in the folded configuration, the handlebar assembly **24** can be positioned generally parallel to the body **20**. The folding assembly **70** includes a release mechanism **72**. The release mechanism **72** can include a handle, button, lever, and/or knob, among mechanical mechanisms. For example, in the illustrated embodiment, the release mechanism **72** includes a handle that, in a first position, locks the rotatable shaft **32** in a substantially upright or riding position and in a second position, allows the rotatable shaft to pivot about a pivot connection **74** (shown in FIG. **13**) between the rotatable shaft **32** and the body **20** such that the rotatable shaft **32** is substantially parallel with a longitudinal axis of the body **20**, as shown in FIG. **11**. In some embodiments, the release mechanism **72** is in wired and/or wireless communication with the controller to receive instructions to release and/or lock the folding assembly **70**. In some embodiments, when the scooter **10** is in the folded configuration, the handlebar assembly is positioned adjacent or near the brake mechanism **96**. In some embodiments, the handlebar assembly **24** abuts and/or touches the brake mechanism **96** and can cause the brake mechanism **96** to contact the rear wheel **30** or otherwise lock the rear wheel **30** to prevent rotation of the rear wheel **30**.

(39) Some embodiments of the electric scooter **10** comprises a battery housing **22** comprising a controller and at least one battery **21**. As shown, the housing **22** can be mounted to the underside of the scooter body **20** underneath the support surface **34**. As illustrated in FIGS. **11** and **12**, the battery **21** is insertable and/or removable from the battery housing **22** when the scooter **10** is in a folded configuration. The battery **21** may be any type of battery, such as a lithium ion rechargeable battery. For example, the battery can have an approximate 1.5-2.5 hour discharge time. As illustrated, the battery **21** can be inserted and/or removed along a direction that is generally parallel with a longitudinal axis of the body **20**.

(40) FIG. **13** illustrates a cross-sectional view of the scooter **10** in the folded configuration. As shown, the battery housing **22** can be located at the forward end of the body **20** of the scooter **10**. This arrangement facilitates easy removable and replacement of the battery **21** when the scooter is in a folded configuration, allowing access to the battery housing **22**. The battery housing can include an aperture on the forward end of the body **20**. The housing **22** can include a closure feature, such as a door (not shown), to close the aperture. In some embodiments, the body **20** and/or the battery **21** comprise a securing feature, such as a detent or hook, configured to secure the battery within the housing **22**. The securing feature can include a release, such as a button or latch, that disengages the securing feature and/or enables the battery **21** to be removed from the housing **22**.

(41) The battery **21** can be configured to be moved into and out of the housing **22**. For example, the battery **21** can be slid into and out of the housing **22** along an axis that is generally parallel with a longitudinal axis of the body **20**. In certain implementations, the battery **21**, when received in the housing **22**, is positioned forward of a front to rear midpoint of the body **20**. This can aid in moving

the center of gravity of the scooter forward, which can provide advantages in riding characteristics. In some variants, the battery **21** is configured to be received in a cavity in a rear of the scooter, such as near the rear wheel **30**. This can increase the amount of weight that is distributed to the rear wheel, which can increase traction of the rear wheel **30**.

(42) The body **20** can include a middle connection piece **23** to provide a point of connection between the support surface **34** and the body **20**, as well as provide additional stability and support. A controller **25** can be located rearward of the battery **21** of the body **20** below the support surface **34**. In some embodiments, the controller **25** is close to the motor **93** and/or between the motor **93** and the battery **21**. In some embodiments, during assembly, the controller **25** may be coupled to the middle connection piece **23**, such as by a clip-in connection, friction fit, fasteners, or otherwise.

(43) As discussed above, some embodiments comprises the motor **93**, such as an electric motor. The motor **93** can be mounted to the underside of the scooter body **20** underneath the support surface **34**, mounted on a rear end of the support surface **34** adjacent the rear drive assembly **80**, mounted on a front end of the support surface **34** adjacent the front wheel **28**, mounted on the handlebar assembly **24** or fork **31**, and/or in any other location. The motor can be operably coupled to the at least one rear wheel **30** to provide powered rotational force upon the at least one rear wheel **30**. In some embodiments, the battery housing **22** and the motor **93** can be located or supported at different locations on the scooter **10**, including on top of the support surface **34** or adjacent either the front or rear wheel **28**, **30**.

(44) In various embodiments, the controller **25** can receive a signal from the throttle assembly **26**. For example, the controller **25** can receive a signal indicative of the amount of speed and/or power to apply to the rear wheel **30**. The controller **25** can provide two-way or one-way transmission to the motor **93**. For example, the controller **25** can instruct the motor **93** to drive the wheel **30** in response to and/or consistent with the signal from the throttle assembly **26**. While control of the scooter **10** can be wireless via the wireless throttle assembly **26**, some variants have wired connections may be provided to connect the throttle, brake, and on/off switch to the motor **93**. Any wired or wireless protocol may be used.

(45) In some embodiments, the scooter **10** includes no wires connecting the battery **21**, controller **25**, and motor **93**. In some embodiments, a conductive material, such as copper, may be incorporated into the body **20** of the scooter **10** once the body **20** has been formed (e.g., injection molded). The conductive material running through the injection molded body **20** acts as a conductor and electrically connects the battery **21**, motor **93**, and controller **25** to control driving operation of the rear wheel **30**. The rear drive assembly **80** may, in some embodiments, be a single module or assembly with the controller and battery separate modules that may be coupled to or inserted within the body **20** during manufacture.

(46) FIGS. **15-21**

(47) FIGS. **15-21** illustrate another embodiment of a scooter **110**. The scooter **110** is similar or identical to the scooter **10** discussed above in many respects. As shown in FIGS. **15-21**, the scooter **110** can include a scooter body **120**, a support surface **134**, a rotatable shaft **132**, and a handlebar assembly **124**, which can be similar to the scooter body, the support surface, the rotatable shaft, and the handlebar assembly discussed above in connection with the scooter **10**. The scooter **110** can include any one, or any combination, of the features of the scooter **10**.

(48) For example, the scooter **110** includes at least one front wheel **128** and at least one rear wheel **130** connected with the scooter body **120**. The front wheel **128** and the rear wheel **130** can be spaced apart from one another with the support surface **134** extending therebetween. The wheels **128**, **130** can be aligned in substantially the same plane, as depicted, and/or located at opposite ends of the support surface **134**. As depicted, some embodiments of the scooter **110** include a rotatable shaft **132** which rotates with the handlebar assembly **124** so that the handlebar assembly **124** can rotate or swivel within a head tube of the scooter body **120**.

(49) In some embodiments, the scooter **110** includes a folding assembly **170** and/or a motor

assembly, such as a rear drive assembly **180**. The folding assembly **170** allows the handlebar assembly **124** and the rotatable shaft **132** to pivot relative to the body **120** such that the handlebar assembly **124** folds against the body **120** for compact transport and storage of the scooter **110**. As shown in FIGS. **15-21**, when the scooter **110** is in the folded configuration, the handlebar assembly **124** can be positioned generally parallel to the body **120**. The folding assembly **170** includes a release mechanism **172**. The release mechanism **172** includes a knob that, in a first position, locks the rotatable shaft **32** in a substantially upright or riding position and in a second position, allows the rotatable shaft to pivot about a pivot connection **174** between the rotatable shaft **132** and the body **120** such that the rotatable shaft **132** is substantially parallel with a longitudinal axis of the body **120**. For example, the release mechanism **172** can be rotated in a first direction to lock the rotatable shaft **132** and a second direction, opposite the first direction, to release the rotatable shaft **132** and allow the rotatable shaft **132** to pivot about the pivot connection **174**.

(50) In some embodiments, the rear drive assembly **180** includes a gear box housing **187**, which can at least partially enclose a gear set. The gear box housing **187** may be part of the main structure of the rear deck or body of the scooter **110**. A portion of the gear set **189** can be enclosed on the outside with a gear box side piece **192**. As shown, the side piece **192** can include a stepped configuration. For example, the stepped configuration can include a first portion **192a** and a second portion **192b**. As shown in the illustrated configuration, the first portion **192a** is substantially flat and the second portion **192b** includes a stepped section that protrudes outwardly from first portion **192a** and is laterally offset from the first portion **192a**. The stepped section can be positioned at least partially adjacent the motor and can extend a portion of the first portion **192a**. The side piece **192** can include one or more apertures **191**, such as to allow airflow from ambient to the motor **193** for cooling. The apertures **191** can be positioned between the first portion **192a** and the second portion **192b** and at least partially surround an end of the motor **193** to allow airflow to directly contact the motor **193** for cooling.

(51) In some embodiments, the rear drive assembly **180** includes a side piece **190** that is removable to provide access to the rear wheel **130** in case the rear wheel should need repair or replacement, such as due to wear, etc. The side piece **190** can be connected to the rear drive assembly **180** to surround at least a portion of the rear drive assembly **180** and can be positioned opposite the side piece **192**, with the gear box housing **187** extending therebetween. As shown in the illustrated embodiment, the side piece **190** can be substantially flat.

(52) As also shown, the motor **193** can be contained in a protective shell, such as a generally cylindrical casing **197**. In some embodiments, the side piece **190** and the casing **197** form a unitary and/or integral unit. In other embodiments, the casing **197** is separate from the side piece **190**. In some embodiments, the casing **197** is formed as part of the motor **193** and includes one or more apertures to allow airflow from ambient to the motor **193** for cooling.

Certain Terminology

(53) Terms of orientation used herein, such as “top,” “bottom,” “horizontal,” “vertical,” “longitudinal,” “lateral,” and “end” are used in the context of the illustrated embodiment. However, the present disclosure should not be limited to the illustrated orientation. Indeed, other orientations are possible and are within the scope of this disclosure. Terms relating to circular shapes as used herein, such as diameter or radius, should be understood not to require perfect circular structures, but rather should be applied to any suitable structure with a cross-sectional region that can be measured from side-to-side. Terms relating to shapes generally, such as “circular” or “cylindrical” or “semi-circular” or “semi-cylindrical” or any related or similar terms, are not required to conform strictly to the mathematical definitions of circles or cylinders or other structures, but can encompass structures that are reasonably close approximations.

(54) Conditional language, such as “can,” “could,” “might,” or “may,” unless specifically stated otherwise, or otherwise understood within the context as used, is generally intended to convey that certain embodiments include or do not include, certain features, elements, and/or steps. Thus, such

conditional language is not generally intended to imply that features, elements, and/or steps are in any way required for one or more embodiments.

(55) Conjunctive language, such as the phrase “at least one of X, Y, and Z,” unless specifically stated otherwise, is otherwise understood with the context as used in general to convey that an item, term, etc. may be either X, Y, or Z. Thus, such conjunctive language is not generally intended to imply that certain embodiments require the presence of at least one of X, at least one of Y, and at least one of Z.

(56) The terms “approximately,” “about,” and “substantially” as used herein represent an amount close to the stated amount that still performs a desired function or achieves a desired result. For example, in some embodiments, as the context may permit, the terms “approximately”, “about”, and “substantially” may refer to an amount that is within less than or equal to 10% of the stated amount. The term “generally” as used herein represents a value, amount, or characteristic that predominantly includes or tends toward a particular value, amount, or characteristic. As an example, in certain embodiments, as the context may permit, the term “generally parallel” can refer to something that departs from exactly parallel by less than or equal to 20 degrees.

(57) Unless otherwise explicitly stated, articles such as “a” or “an” should generally be interpreted to include one or more described items. Accordingly, phrases such as “a device configured to” are intended to include one or more recited devices. Such one or more recited devices can also be collectively configured to carry out the stated recitations. For example, “a processor configured to carry out recitations A, B, and C” can include a first processor configured to carry out recitation A working in conjunction with a second processor configured to carry out recitations B and C.

(58) The terms “comprising,” “including,” “having,” and the like are synonymous and are used inclusively, in an open-ended fashion, and do not exclude additional elements, features, acts, operations, and so forth. Likewise, the terms “some,” “certain,” and the like are synonymous and are used in an open-ended fashion. Also, the term “or” is used in its inclusive sense (and not in its exclusive sense) so that when used, for example, to connect a list of elements, the term “or” means one, some, or all of the elements in the list.

(59) Overall, the language of the claims is to be interpreted broadly based on the language employed in the claims. The language of the claims is not to be limited to the non-exclusive embodiments and examples that are illustrated and described in this disclosure, or that are discussed during the prosecution of the application.

SUMMARY

(60) Although this invention has been disclosed in the context of certain embodiments and examples, it will be understood by those skilled in the art that the present disclosure extends beyond the specifically disclosed embodiments to other alternative embodiments and/or uses of the invention and obvious modifications and equivalents thereof. In particular, while the present systems and methods have been described in the context of particular embodiments, the skilled artisan will appreciate, in view of the present disclosure, that certain advantages, features and aspects of the systems and methods may be realized in a variety of other applications, many of which have been noted above. Various features and aspects of the disclosed embodiments can be combined with or substituted for one another in order to form varying modes of the conveyor. The scope of this disclosure should not be limited by the particular disclosed embodiments described herein.

(61) Additionally, it is contemplated that various aspects and features of the invention described can be practiced separately, combined together, or substituted for one another, and that a variety of combination and subcombinations of the features and aspects can be made and still fall within the scope of the invention. Certain features that are described in this disclosure in the context of separate implementations can also be implemented in combination in a single implementation. Conversely, various features that are described in the context of a single implementation can also be implemented in multiple implementations separately or in any suitable subcombination.

Although features may be described above as acting in certain combinations, one or more features from a claimed combination can, in some cases, be excised from the combination, and the combination may be claimed as any subcombination or variation of any subcombination.

(62) Moreover, while operations may be depicted in the drawings or described in the specification in a particular order, such operations need not be performed in the particular order shown or in sequential order, and all operations need not be performed, to achieve the desirable results. Other operations that are not depicted or described can be incorporated in the example methods and processes. For example, one or more additional operations can be performed before, after, simultaneously, or between any of the described operations. Further, the operations may be rearranged or reordered in other implementations. Also, the separation of various system components in the implementations described above should not be understood as requiring such separation in all implementations, and it should be understood that the described components and systems can generally be integrated together in a single product or packaged into multiple products. Additionally, other implementations are within the scope of this disclosure.

(63) Some embodiments have been described in connection with the accompanying drawings. The figures are drawn to scale, but such scale should not be limiting, since dimensions and proportions other than what are shown are contemplated and are within the scope of the disclosed invention. Distances, angles, etc. are merely illustrative and do not necessarily bear an exact relationship to actual dimensions and layout of the devices illustrated. Components can be added, removed, and/or rearranged. Further, the disclosure herein of any particular feature, aspect, method, property, characteristic, quality, attribute, element, or the like in connection with various embodiments can be used in all other embodiments set forth herein. Additionally, any methods described herein may be practiced using any device suitable for performing the recited steps.

(64) In summary, various embodiments and examples of personal mobility devices, such as scooters, have been disclosed. Although the devices have been disclosed in the context of those embodiments and examples, this disclosure extends beyond the specifically disclosed embodiments to other alternative embodiments and/or other uses of the embodiments, as well as to certain modifications and equivalents thereof. This disclosure expressly contemplates that various features and aspects of the disclosed embodiments can be combined with, or substituted for, one another. Thus, the scope of this disclosure should not be limited by the particular embodiments described above, but should be determined only by a fair reading of the claims that follow.

Claims

1. A two-wheel scooter comprising: a body comprising a deck, a front end, a rear end, and a longitudinal axis, the deck configured to support a user; a steering assembly connected to the front end of the body, the steering assembly comprising a handlebar, a steering shaft, and a front wheel; a rear wheel connected to the rear end of the body; a rear drive assembly configured to drive the rear wheel, the rear drive assembly comprising an electric motor and a transmission; a battery housing under the deck, the battery housing comprising an aperture; and a battery in the battery housing, the battery configured to provide electric power to the electric motor, the battery selectively removable from the battery housing through the aperture along a direction that is generally parallel with the longitudinal axis of the body, wherein when the battery is in the battery housing a front end of the battery protrudes longitudinally frontward from the front end of the body, thereby providing a grasping portion to aid removal of the battery.

2. The scooter of claim 1, further comprising a folding assembly that connects the steering assembly and the body, the folding assembly comprising a hinge and a lock, the folding assembly configured to enable the steering assembly to pivot relative to the body between an unfolded state and a folded state, wherein: in the unfolded state, the steering assembly blocks removal of the battery from the battery housing; and in the folded state, the steering assembly does not block

removal of the battery from the battery housing.

3. The scooter of claim 2, wherein, in the unfolded state and with the front and rear wheels on a riding surface, the battery is directly below the folding assembly.

4. The scooter of claim 1, wherein the direction is toward the front end of the body.

5. The scooter of claim 1, wherein the aperture is on a front end of the battery housing.

6. The scooter of claim 1, wherein the battery housing is connected to the front end of the body.

7. The scooter of claim 1, wherein the rear drive assembly is configured to drive the rear wheel in a first rotational direction and is configured to allow the rear wheel to rotate freely in a second rotational direction opposite of the first rotational direction.

8. A method of replacing a spent battery in an electric scooter, the method comprising: disengaging a lock of a folding assembly of the scooter; folding the scooter, wherein folding the scooter comprises: pivoting in a first rotational direction the steering assembly of the scooter relative to a body of the scooter; and activating, with the steering assembly, a brake of the scooter against a rear wheel of the scooter; withdrawing the spent battery from the body in a first linear direction generally parallel to a longitudinal axis of the body; inserting a fresh battery into the body in a second linear direction generally parallel to the longitudinal axis; unfolding the scooter, wherein unfolding the scooter comprises pivoting in a second rotational direction the steering assembly relative to the body; and engaging the lock of the folding assembly.

9. The method of claim 8, wherein the first linear direction is toward a front of the scooter and the second linear direction is toward a rear of the scooter.

10. The method of claim 8, wherein folding the scooter further comprises moving a front wheel of the scooter out of a position that blocks withdrawal of the spent battery, and wherein unfolding the scooter further comprises moving the front wheel into a position that blocks withdrawal of the fresh battery.

11. The method of claim 8, wherein withdrawing the spent battery from the body comprises grasping a front of the battery that protrudes from the body of the scooter.

12. A two-wheel scooter comprising: a body comprising a deck, a front end, a rear end, and a longitudinal axis, the deck configured to support a user; a steering assembly connected to the front end of the body, the steering assembly comprising a handlebar, a steering shaft, and a front wheel; a rear wheel connected to the rear end of the body; a rear drive assembly configured to drive the rear wheel, the rear drive assembly comprising an electric motor and a transmission; a battery configured to provide electric power to the electric motor; a brake configured to move into contact with the rear wheel; and a folding assembly configured to enable the steering assembly to pivot relative to the body between an unfolded state and a folded state, wherein in the folded state the steering assembly contacts the brake, thereby inhibiting rotation of the rear wheel.

13. The scooter of claim 12, wherein the folding assembly comprises a hinge and a lock.

14. The scooter of claim 13, wherein the lock comprises a rotatable knob.

15. The scooter of claim 12, wherein the battery is selectively removable from the body along a direction that is generally parallel with the longitudinal axis of the body.

16. The scooter of claim 12, wherein the brake is configured to be activated by a rider's foot.

17. The scooter of claim 12, wherein the rear drive assembly is configured to drive the rear wheel in a first rotational direction and is configured to allow the rear wheel to rotate freely in a second rotational direction opposite of the first rotational direction.

18. A two-wheel scooter comprising: a body comprising a deck, a front end, a rear end, and a longitudinal axis, the deck configured to support a user; a steering assembly connected to the front end of the body, the steering assembly comprising a handlebar, a steering shaft, and a front wheel; a rear wheel connected to the rear end of the body; a rear drive assembly configured to drive the rear wheel, the rear drive assembly comprising an electric motor and a transmission; a battery housing under the deck, the battery housing comprising an aperture; a battery in the battery housing, the battery configured to provide electric power to the electric motor, the battery selectively removable

from the battery housing through the aperture along a direction that is generally parallel with the longitudinal axis of the body; and a folding assembly that connects the steering assembly and the body, the folding assembly comprising a hinge and a lock, the folding assembly configured to enable the steering assembly to pivot relative to the body between an unfolded state and a folded state, wherein: in the unfolded state and with the front and rear wheels on a riding surface, the steering assembly blocks removal of the battery from the battery housing and the battery is directly below the folding assembly; and in the folded state, the steering assembly does not block removal of the battery from the battery housing.

19. The scooter of claim 18, wherein the direction is toward the front end of the body.

20. The scooter of claim 18, wherein the aperture is on a front end of the battery housing.

21. The scooter of claim 18, wherein the battery housing is connected to the front end of the body.

22. The scooter of claim 18, wherein the rear drive assembly is configured to drive the rear wheel in a first rotational direction and is configured to allow the rear wheel to rotate freely in a second rotational direction opposite of the first rotational direction.
